AUTONOMOUS MULTITASKING RESCUE ROBOT IN UNMANNED ZONES

A MAJOR PROJECT PHASE II REPORT

Submitted in partial fulfillment of the requirement for the award of the degree

Bachelor of Engineering

In

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Submitted By

DEEPALI DINESH B	**1MJ22EC042**
GANAVI V R	**1MJ21EC051**
GEETHA M K	**1MJ21EC054**
HAMSA K	**1MJ22EC059**

Under the Guidance of
Dr. Chippy M Manu
Assistant Professor
Department of Electronics and Communication Engineering, MVJCE

MVJ College of Engineering, Bengaluru
(An Autonomous Institute)
Affiliated to VTU, Belagavi, Approved by AICTE, New Delhi
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MVJ COLLEGE OF ENGINEERING, BENGALURU-560067

(Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

Certified that the Major Project Phase II work titled **'AUTONOMOUS MULTI TASKING RESCUE ROBOT IN UNMANNED ZONES'** is carried out by **DEEPALI DINESH B (1MJ22EC042), GANAVI V R (1MJ22EC051), GEETHA M K (1MJ22EC054), HAMSA K (1MJ22EC059)** who are Bonafide students of MVJ College of Engineering, Bengaluru, in partial fulfillment for the award of degree of **Bachelor of Engineering in Electronics and Communication Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2025-2026. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project phase II report deposited in the departmental library. The major project report II has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said Degree.

Signature of Guide	**Signature of Head of the Department**	**Signature of Dean**
Dr. Chippy M Manu Assistant Professor Dept. of ECE MVJ College of Engineering Bengaluru	**Dr. Shima Ramesh Maniyath** Head of the Department Dept. of ECE MVJ College of Engineering Bengaluru	**Dr. M Brindha** Dean School of Electrical Science MVJ College of Engineering Bengaluru

External Viva

Name of Examiners	**Signature with Date**
1	
2	

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DECLARATION

We, DEEPALI DINESH B (1MJ22EC042), GANAVI V R (1MJ22EC051), GEETHA M K (1MJ22EC054), HAMSA K (1MJ22EC059), students of 8th semester B.E., Department of Electronics and Communication Engineering, MVJ College of Engineering, Bengaluru, hereby declare that the major project phase II titled '**AUTONOMOUS MULTI TASKING RESCUE ROBOT IN UNMANNED ZONES**' has been carried out by us and submitted in partial fulfilment for the award of Degree of **Bachelor of Engineering** in **Electronics and Communication** during the year 2025-2026.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare that any Intellectual Property Rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru and we will be one of the authors of the same.

Place: Bengaluru

Date:

Name **Signature**

1. **DEEPALI DINESH B(1MJ22EC042)**
2. **GANAVI V R(1MJ22EC051)**
3. **GEETHA M K(1MJ22EC054)**
4. **HAMSA K(1MJ22EC059)**

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ABSTRACT

In disaster-stricken environments such as fire zones, gas leaks, and building collapses, human intervention for rescue operations can be highly dangerous and life-threatening. To address this challenge, we propose a Multitasking Rescue Robot capable of detecting environmental hazards, locating victims, assessing their health condition, and transmitting real-time data to a remote operator. The system is built using an Arduino Mega microcontroller and integrates multiple sensors including fire, gas, metal, ultrasonic, heartbeat, and temperature sensors to ensure comprehensive environmental and health monitoring.

The robot is equipped with a pick and place mechanism using servo-controlled arms for safe handling of injured individuals. It navigates autonomously with the help of an ultrasonic sensor and DC motors controlled via an H-Bridge, allowing it to move through complex terrains. Bluetooth communication enables manual override and control, while Zigbee is used for long-range wireless transmission of sensor data to a laptop for monitoring.

To enhance decision-making, the vital signs of the rescued person (heartbeat and temperature) are sent to a machine learning module hosted on the laptop. The model analyzes the data and predicts the person's health condition, classifying it as normal, mild risk, or critical, thus allowing rescuers to prioritize actions accordingly.

This robot serves as a powerful tool in modern rescue missions, minimizing human risk while maximizing efficiency and accuracy in emergency response scenarios.

TABLE OF CONTENTS		Page No.
	CERTIFICATE	I
	DECLARATION	II
	ACKNOWLEDGEMENT	III
	ABSTRACT	IV
CHAPTER 1	**INTRODUCTION**	9
	1.1 Need for a Multitasking Rescue Robot	11
	1.2 Disaster Challenges and Technological Opportunities	11
	1.3 Foundation Technologies Used in this Project	12
	1.4 Importance of Multitasking in Modern Rescue Robotics	14
	1.5 Applications of the Multitasking Rescue Robot	14
	1.6 Motivation for the Project	15
	1.7 Summary of the Robot's Capabilities	15
	1.8 Conclusion of Extended Introduction	16
CHAPTER 2	**LITERATURE SURVEY**	17
CHAPTER 3	**EXISTING AND PROPOSED SYTEM**	19
	3.1 Existing System	20
	3.2 Proposed System	21
CHAPTER 4	**METHODOLOGY**	22
	4.1 METHODOLOGY	23
	4.1.1 System Initialization	23

	4.1.2 Autonomous Navigation	24
	4.1.3 Environmental Hazard Detection	24
	4.1.4 Victim Detection & Health Monitoring	24
	4.1.5 Machine Learning–Based Health Prediction	25
	4.2 SYSTEM REQUIREMENTS	25
	4.2.1 Hardware Requirements	26
	4.2.2 Software Requirements	27
CHAPTER 5	**IMPLEMENTATION**	28
	5.1 System Design	29
	5.2 Block Diagram	32
CHAPTER 6	**RESULTS & DISCUSSIONS**	35
	6.1 Overview of Experimental Setup	36
	6.2 Practical Implications	41
CHAPTER 7	**CONCLUSIONS & FUTURE SCOPE**	42
	7.1 Conclusion	43
	7.2 Future Scope	44
REFERENCES		46

LIST OF FIGURES

S. NO.	NAME OF FIGURE	PAGE NO
5.1	Flow Chart of the System	29
5.2	Block Diagram	32
6.1	Sensor Integration	37
6.2	Robotic Arm	42

CHAPTER 1

INTRODUCTION

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INTRODUCTION

Disasters, either natural or artificial, are unexpected, destructive, and often catastrophic to human life and infrastructure. Earthquakes, building collapses, industrial explosions, toxic gas leaks, landslides, forest fires, and war-zone incidents are among the most common crisis events where survivors fall into urgent need of assistance. In these extreme circumstances, the environment at the disaster site is usually extremely unstable, full of debris, smoke, heat, toxic chemicals, or hazardous materials, making it very dangerous for human rescue workers to operate safely.

Traditional search-and-rescue operations require specially trained personnel to go into the disaster area and physically locate survivors, assess risk, and administer first aid. Even with the purchase of specialized equipment and training by rescue organizations around the world, these methods continue to be slow, dangerous, and very labor-intensive, particularly when the environment presents unseen hazards like sudden collapse, explosive gases, and flash fire. Every minute wasted in finding and stabilizing a victim minimizes their survival rate.

New frontiers have opened up for minimizing risks in hazardous environments with the evolution of robotics and sensor-based intelligent systems. In the last decade, robotic platforms have been explored increasingly for their applications in disaster detection, environmental monitoring, unmanned inspections, and emergency support. Robots can enter areas inaccessible to humans, gather crucial information, and operate continuously without fatigue or fear. These advantages make them ideal tools for modern search-and-rescue operations.

Most existing rescue robots tend to focus on single functionalities, for example, fire detection, gas leakage identification, or simple obstacle avoidance. Very few integrate all the critical rescue functions into one cost-effective and compact platform. This gap in the field necessitates the development of a multi-functional, intelligent, sensor-integrated robot capable of handling many rescue tasks simultaneously.

1.1 Need for a Multitasking Rescue Robot

A rescue robot deployed during an emergency must possess the following characteristics:

- Reliable enough to withstand harsh conditions
- Able to detect hazards and make navigation decisions intelligently
- Connected enough to send real-time data to a control center in another location
- Equipped with biomedical sensing to assess victims
- Able to interact with the environment physically

The Multitasking Rescue Robot reported herein is developed in consideration of these critical needs. It is an integrated system comprising:

- Fire detection
- Gas concentration detection
- Metal detection for hidden hazards
- Obstacle avoidance and autonomous navigation
- Victim health monitoring (heartbeat + temperature)
- Wireless data transmission using Zigbee
- Short-range manual control by Bluetooth
- Pick-and-place mechanism for assisting victims

In addition, physiological readings are sent from the robot to a remote computer, where a machine learning model analyzes the condition of the victim and classifies it as Normal, Mild, or Critical. This capability gives an important decision-support mechanism to the rescue teams for them to prioritize activities with real-time health insights

1.2 Disaster Challenges and Technological Opportunities

A multitasking rescue robot addresses unpredictable and dangerous disaster environments where human access is limited.

1.2.1 Environmental Challenges in Rescue Missions

When a disaster strikes, many environmental issues hamper rescue work:

low visibility due to dust, smoke or darkness

Unstable debris posing risks of collapse

Toxic gases, such as CO, LPG, and methane, are dangerous to rescuers.

High temperatures from fire outbreaks

Narrow or blocked pathways

Unknown obstacles or metallic hazards

These challenges considerably delay the detection of victims. Victims who are experiencing low heart rates, hypothermia, trauma, or internal injuries can rapidly deteriorate without immediate medical analysis before rescuers reach them.

1.2.2 Advantages of Robotic Intervention

Robots have critical advantages such as:

Entering dangerous zones without risking human life

Continuous monitoring of environmental parameters

Precision in the sensor readings: temperature, heart rate, gas concentration

Efficient data transmission

Ability to repeat tasks without fatigue

Higher endurance in harsh environments

The proposed robot leverages technological advances in embedded systems, sensor engineering, microcontrollers, and machine learning to address multi-dimensional challenges in real-world rescue missions.

1.3 Foundation Technologies Used in this Project

A multitasking rescue robot uses different sensors, smart control systems, and wireless communication to detect dangers and locate victims. It moves on its own, avoids obstacles, and sends live updates to the rescue team. Machine learning helps the robot understand the situation and choose which victim needs help first.

1.3.1 Arduino Mega 2560 as the Control Unit

The Arduino Mega 2560 is used because:

- High number of digital/analog pins
- Stable processor (ATmega2560)
- Support for multiple simultaneous sensor interfaces
- Cost-effectiveness
- Reliability in embedded applications
- It serves as the central processing unit, dealing with:
- Sensor data acquisition
- Motor control

- Wireless module communication
- updating of LCD display
- Decision logic for navigation

1.3.2 Sensors for Environmental Monitoring

- The robot integrates diverse sensors:
- Fire Sensor\Detects flame/heat sources
- MQ series gas sensor Identifies LPG, CO, methane & toxic gas leaks
- Metal Sensor: It detects metallic debris or mines.
- Ultrasonic Sensor: Obstacle detection & distance measurement

These sensors allow the robot to build a rich model of the environment and to identify potential hazards well in advance.

1.3.3 Victim Health Monitoring

Two important biomedical sensors are utilized:

- Heartbeat Sensor: Photoplethysmography-based pulse detection
- DHT11/DS18B20 Temperature Sensor: This measures the skin temperature of the victim.

This multi-sensor system can estimate the physiological condition of the victim.

1.3.4 Wireless Communication Modules

The robot uses:

Bluetooth HC-05 for short-range manual control

Zigbee TB387 for long-range, low-power data transmission

It finds applications in SAR missions on account of its high reliability, mesh capability, and extended range (~500 m in open environments).

1.3.5 Machine Learning-Based Decision Support

A machine learning model is implemented to classify the condition of the victim using Python via Jupiter Notebook. It analyzes:

- Heartbeat values
- Temperature readings

Based on data, it predicts:

- Normal
- Mild Condition -Requires Attention
- Critical condition: needs immediate rescue

This automated triage helps rescue teams to act swiftly and efficiently.

1.4 Importance of Multitasking in Modern Rescue Robotics

Traditionally, robots for SAR have been specialized. Some detect only fire or gas; others focus solely on obstacle navigation or thermal imaging. A multitasking robot is a major advance because it brings all the necessary rescue functions into one integrated machine.

1.4.1 Advantages of Multitasking Capabilities

- Faster victim detection
- Rapid risk assessment
- On-site medical evaluation
- Real-time communication
- Reduced requirement for many robots
- Lower operational costs

By integrating navigation, sensing, mechanical action, and medical prediction, the robot fundamentally enhances rescue efficiency.

1.5 Applications of the Multitasking Rescue Robot

Applications of the designed robot include:

- Earthquake rescue scenarios
- Industrial accidents and chemical plant monitoring
- Fire outbreaks in buildings
- Gas leakage incidents
- War zone casualty assistance
- Exploration of hazardous tunnels or mines
- Disaster-prone remote environments

Additionally, this can be adapted for:

Smart surveillance

1.6 Motivation for the Project

The motivation behind developing this robot includes:

1. The increasing casualty numbers in disasters due to late detection and lack of medical assessment
2. There is a need for safe, quick, and reliable systems that minimize risk to human rescuers.
3. Opportunities to utilize Arduino-based solutions to obtain low-cost yet powerful rescue equipment
4. Integration of machine learning to support real-time decision-making
5. Interest in developing a scalable platform for future robotic enhancements including AI vision, GPS, and autonomous mapping.

This project integrates Embedded Systems, Robotics, Wireless Communication, and Intelligent Decision-Making into the design and development of a practical and effective rescue solution.

1.8 Conclusion of Extended Introduction

A multitasking rescue robot represents an important step toward safer, faster, and smarter disaster response operations. In that respect, the integration of multi-sensor environmental monitoring, biomedical data acquisition, autonomous movement, wireless communication, and machine learning merges into a holistic approach toward modern rescue robotics. The developed system overcomes the weaknesses of the traditional rescue methods and existing single-function robots with an all-in-one platform capable of operating under dangerous conditions.

Chapter Summary

The proposed multitasking rescue robot can move on its own while avoiding obstacles and detect fire, gas, and metallic objects in the environment. It can sense victims and measure their heartbeat and temperature for quick assessment. The robot can also lift or shift small debris using its pick-and-place mechanism and sends real-time data to rescuers through Zigbee and Bluetooth. Predict the victim's health status using ML algorithms and Display information on an LCD for local assessment, That makes the tool highly versatile in emergency response situations.

CHAPTER 2

LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

1. According to Zhihao Chen et.al[1], it implements framework for object identification, localization and monitoring for smart mobility applications such as road traffic and railway climate. An object detection and tracking approach was firstly carried out within two deep learning approaches: You Only Look Once (YOLO) V3 and Single Shot Detector (SSD).

2. Zhong-Qiu Zhao et.al[2], a analysis of deep learning focused on the frameworks for object detection is presented in this paper. Generic object detection architectures are addressed in context with convolution neural network (CNN), along with some modifications and useful tricks to boost detection efficiency.

3. Licheng Jiao et.al [3], This paper highlights the rapid growth of deep learning networks for detection tasks, the efficiency of object detectors that has been greatly enhanced.

4. Yakup Demir2 et.al[4], addresses autonomous driving that involves reliable and accurate detection and identification in real drivable environments of surrounding objects. While numerous algorithms for object detection have been proposed, not all are robust enough to detect and identify occluded or truncated objects. A new hybrid Local Multiple System (LMCNNSVM) based on Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs) is proposed in this paper due to its powerful extraction capability and robust classification property.

5. Mukesh Tiwari ed.al[5] discusses that the identification and tracking of objects are important research areas due to daily change in object motion and variance in scene size, occlusions, variations in appearance, and changes in ego-motion and illumination. Specifically, selection of features is a vital part of tracking objects.

Shwetha R and Dr. Chethan H K has done a work on designing an economical robot, which works using AVR, MCU, PIR sensor. This robot senses the human body temperature using PIR sensor and an alarm/indicator is used to indicate the signal when it detects alive body and this message is sent through SMS using GSM technology to enable rescue operation. λ

Mauricio Correa Et al. has given the concept of enabling robots to detect and identify humans in domestic environment. This work was done with the aid of Thermal and Visual Information sources that were integrated to detect humans and further processed to verify it.

Wasif Naeem Et al. has presented an initial study of the autopilot development of an unmanned surface vehicle (USV). The USV named Springer is being developed to carry out pollutant tracking, and environmental and hydrographic surveys in rivers, reservoirs, inland waterways and coastal waters, particularly where shallow waters prevail. The catamaran shaped autonomous vessel is modelled as a two input, single output system whereas the autopilot selected is a genetic algorithm based model predictive controller (MPC). The USV dynamic model is obtained using system Identification techniques. Simulation results depicting the autopilot performance in response to changes in the desired course are demonstrated.

Usha Tiwari Et al. has explained about designing a Robot to navigate in the rubble with various sensors. This method used 2 methods to detect alive human, one is IR radiation emerging from the live humans and other is using the sound or cry for the help made from the humans.

Amon Tunwannarux Et al. has presented a rescue robot .It has a double track robot system, it is good for unstructured landscape and able to climb over the pile of collapse. This robot is equipped with a lot of sensors such as IR temperature sensors, distance sensors, odometer dsensors,pitch/roll,compass sensor, three pan/tilt CCD cameras and Voice sensor. The double track mechanism of the robot wheels and information provided about various sensors are mainly referred.

Yogesh V. Bangalkar and S. M. Kharad has presented a prototype robot which is extensively useful for military applications. PIR sensor is used to detect human. A Passive Infra-Red sensor is an electronic device which measures infrared light radiating from objects in its field of interpretation. Seeming motion is detected when an infrared source with one temperature, such as human, passes in noticeable of an infrared source with another temperature, it detects. It acts as a motion finder. Concepts like GPS tracking mechanism, Forward and reverse direction movement using geared motors are also discussed in this paper.

CHAPTER 3

EXISTING AND PROPOSED SYSTEM

CHAPTER 3

EXISTING AND PROPOSED SYSTEM

In the existing rescue systems, most robots are designed to perform only one or two specific tasks such as obstacle detection, fire sensing, or basic navigation. These systems usually rely on manual control, single-sensor operation, and limited communication features.

3.1 Existing System

Several robotic and sensor-based rescue systems have been developed to assist in disaster response and emergency operations. These existing systems mainly focus on individual functionalities such as fire detection, victim locating, or environmental monitoring but often lack integration of multiple capabilities in a single platform.

1. Fire Detection Robots: Many rescue robots are equipped with fire sensors and cameras to navigate hazardous areas and identify fire sources. These robots can relay the fire location to rescue teams, but their functionality is usually limited to fire detection and suppression support without victim health monitoring or environmental hazard detection.

2. Gas Leak Detection Robots: Robots equipped with gas sensors are used in chemical plants or industrial sites to detect toxic or combustible gases. While effective for environmental hazard detection, these robots typically lack mobility suited for rough terrain and do not incorporate victim detection or health monitoring.

3. Victim Detection and Monitoring Robots: Some systems focus on locating victims trapped under rubble using thermal imaging or sound detection and may include vital sign monitoring sensors. However, these systems often lack the ability to navigate autonomously or handle physical rescue tasks like moving debris.

4. Wireless Communication in Rescue Robots: Several projects employ wireless modules such as Bluetooth, Wi-Fi, or Zigbee for remote control and data transmission. However, most systems use a single communication method and do not combine short-range and long-range wireless technologies for enhanced connectivity.

5. Use of Machine Learning in Rescue Operations: Emerging systems are beginning to integrate machine learning algorithms to analyze sensor data for predicting the health status of victims or identifying hazards.

Overall, while existing systems excel in specific tasks, there is a gap in creating a multifunctional rescue robot that combines autonomous navigation, multi-hazard detection, victim health monitoring, mechanical rescue capabilities, wireless communication, and real-time health status prediction using machine learning. The proposed Multitasking Rescue Robot aims to fill this gap by integrating all these features into a single, cost-effective, and portable platform using the Arduino Mega microcontroller.

3.2 Proposed System

The proposed Multitasking Rescue Robot is designed to address the limitations of existing rescue systems by integrating multiple sensing, communication, and mechanical functionalities into a single cohesive platform controlled by an Arduino Mega microcontroller. The system combines environmental hazard detection, victim health monitoring, autonomous navigation, and mechanical assistance with real-time wireless data transmission and intelligent health prediction. The robot is equipped with an array of sensors including fire, gas, and metal sensors to detect dangerous environmental conditions such as flames, toxic gases, and metallic debris or obstacles. Ultrasonic sensors provide obstacle detection and enable autonomous navigation, allowing the robot to traverse complex and hazardous terrains without human intervention. A pick-and-place robotic arm, driven by DC motors and controlled via an H-Bridge driver, enables the robot to lift light debris or assist in relocating trapped individuals, thereby expanding its role beyond just detection.

Chapter Summary

To monitor the health of trapped victims, heartbeat and temperature sensors are incorporated, gathering vital physiological data. This information is transmitted wirelessly to a remote laptop or control center using a dual communication system: Bluetooth for short-range control and Zigbee for long-range, reliable data transfer. The transmitted data is then processed by a machine learning module that analyzes the victim's condition and predicts health status categories such as normal, mild, or critical, assisting rescue teams in prioritizing their efforts effectively.

CHAPTER 4

METHODOLOGY

CHAPTER 4

METHODOLOGY

The methodology involves integrating multiple sensors to detect fire, gas, metal objects, and vital signs of victims. The robot uses an embedded controller to process sensor data and navigate autonomously while avoiding obstacles. A pick-and-place mechanism helps the robot lift or move small debris during rescue tasks. Machine learning algorithms analyze the collected data to assess victim condition and prioritize actions. Finally, all real-time information is transmitted wirelessly to the rescue team using Zigbee and Bluetooth for faster decision-making

4.1 Methodology

The design a multi tasking Rescue Robot It details the sequential procedures followed to achieve autonomous navigation, multi-sensor integration, hazard detection, and victim health monitoring. This section explains the operational flow beginning from system initialization to data acquisition, wireless transmission, and machine-learning-based health classification. Each hardware and software component is coordinated to ensure accurate sensing, reliable communication, and safe mechanical assistance during rescue operations.

4.1.1 System Initialization

- When the robot is powered ON, the Arduino Mega configures all I/O pins and initializes each connected sensor, including fire, gas, metal, ultrasonic, heartbeat, and temperature sensors.
- Communication modules **Bluetooth (HC-05)** and **Zigbee (2.4 GHz)** are activated to establish communication readiness.
- Internal timers, interrupts, motor drivers (L293D), and servo controllers are set to their default operational states.
- The **16×2 LCD** displays boot information such as *"System Ready," "Sensors Initializing,"* ensuring transparency during startup

4.1.2 Autonomous Navigation

- The **ultrasonic sensor (HC-SR04)** continuously emits and receives sound pulses to measure distance from nearby obstacles.
- Real-time distance data is processed by the microcontroller to determine safe movement directions
- The Arduino controls **DC motors** through the L293D H-Bridge, adjusting direction and speed based on surrounding obstacles.
- Navigation logic helps the robot move safely across disaster areas such as collapsed buildings, debris-filled paths, or low-visibility zones.
- If an obstacle is too close, the robot automatically performs corrective actions (e.g., stop, reverse, turn)

4.1.3 Environmental Hazard Detection

- The **fire/flame sensor** continuously checks for high IR radiation or flame signatures.
- The **gas sensor (MQ series)** monitors toxic gases, smoke levels, or combustible chemical leaks.
- The **metal sensor** detects the presence of metallic objects, indicating structural hazards, buried materials, or trapped equipment.
- Whenever any sensor detects danger beyond predefined thresholds:
    - The LCD displays an alert message (e.g., “Fire Detected,” “Gas Leak Warning”).
    - A wireless alert packet is sent via Zigbee to the monitoring station.
    - A buzzer may activate to indicate high-risk scenarios.

4.1.4 Victim Detection & Health Monitoring

- Once the robot identifies a potential victim (either manually by the operator or by proximity sensing), it activates the biomedical sensing mode.
- The **heartbeat sensor** measures pulse rate using reflection-based optical sensing.

- The **temperature sensor (DS**
- **18B20 or DHT11)** records the victim's body temperature.
- Both readings are verified, processed, and immediately shown on the LCD for on-site interpretation.
- These health parameters are also transmitted wirelessly using the Zigbee module to the remote control unit for further analysis.

4.1.5 Machine Learning–Based Health Prediction

 The transmitted biomedical data is received by a laptop running a Python-based machine learning model.

 Preprocessing steps (normalization, scaling, or threshold mapping) prepare the data for prediction.

 The trained ML classifier evaluates the received values and categorizes the victim's health condition as:

- Normal
- Mild Risk
- Critical

 The prediction result is shown at the control center, helping rescue teams prioritize high-risk victims immediately.

 This intelligent system improves decision accuracy, especially when medical personnel are not immediately available

4.2 SYSTEM REQUIREMENTS

The system requirements define all the essential hardware and software components needed to design and operate the multitasking rescue robot effectively. These requirements ensure that the robot can perform multiple rescue operations—such as sensing, navigation, communication, and decision-making—smoothly and reliably. By specifying sensors, controllers, power sources, communication modules, and processing units, the system requirements provide a clear foundation for building a robust and efficient rescue platform.

4.2.1 Hardware Requirements:

Microcontroller and Processing Units

- Arduino Mega 2560 – Main controller for sensor integration, motor control, and communication handling.
- Motor Driver (L293D H-Bridge) – Enables bidirectional control of DC motors.

Sensors

- Ultrasonic Sensor (HC-SR04) – Obstacle detection and autonomous navigation.
- Fire/Flame Sensor – Detects flames and high-temperature IR radiation.
- Gas Sensor (MQ Series) – Identifies toxic gases and combustible chemicals.
- Metal Detector Sensor – Detects metallic hazards or hidden objects.
- Heartbeat (Pulse) Sensor – Measures victim's pulse rate.
- Temperature Sensor (DS18B20 / DHT11) – Records victim's body temperature.

Communication Modules

- Bluetooth Module (HC-05) – Short-range manual control and data exchange.
- Zigbee Module (2.4 GHz) – Long-range wireless data transmission for real-time monitoring.

Actuators

- DC Motors – Used for wheel movement and robot locomotion.
- Servo Motors – Operate the pick-and-place robotic arm.
- Buzzer – Warning alerts for hazards.

Display and Interfaces

- 16×2 LCD Display – Shows sensor readings and system status.
- Power Supply/Battery – Provides stable 5V/12V power for electronics and motors.
- Relay Module – For switching high-power components if required.
- Chassis/Robot Frame – Structure for mounting all components.

4.2.2 Software Requirements:

- Arduino IDE – Writing, compiling, and uploading Embedded C programs to the Arduino Mega.
- Embedded C/C++ – Main programming language for controlling sensors and actuators.

Machine Learning & Analysis Tools

- Python – For data processing and ML model development.
- Jupyter Notebook / Anaconda Navigator – For training and running machine-learning models.

Python Libraries

- NumPy – Numerical operations.
- Pandas – Data handling and preprocessing.
- Scikit-learn – Machine learning model training and prediction.
- Matplotlib / Seaborn – Visualization of datasets and model outputs.
- OpenCV (Optional) – For future upgrades involving video or image processing

Chapter Summary

The methodology involves integrating multiple sensors to detect fire, gas, metal objects, and vital signs of victims. The robot uses an embedded controller to process sensor data and navigate autonomously while avoiding obstacles. A pick-and-place mechanism helps the robot lift or move small debris during rescue tasks. Machine learning algorithms analyze the collected data to assess victim condition and prioritize actions. Finally, all real-time information is transmitted wirelessly to the rescue team using Zigbee and Bluetooth for faster decision-making.

CHAPTER 5
IMPLEMENTATION

CHAPTER 5
IMPLEMENTATION

The implementation phase describes how the proposed multitasking rescue robot is developed by integrating all hardware and software components into a functional system. This section explains the step-by-step process of assembling sensors, microcontrollers, communication modules, and mechanical parts, followed by programming the robot to perform autonomous navigation, environmental monitoring, and real-time data transmission.

5.1 System Design

The robot uses an Arduino Mega to control multiple environmental and biomedical sensors, motor drivers, and communication modules. Autonomous navigation is achieved through ultrasonic sensing, while fire, gas, and metal sensors provide hazard detection. Wireless communication is enabled using Bluetooth and Zigbee, allowing real-time data transmission to a remote monitoring system. A machine learning model processes the vital-sign data to classify victim condition and support rescue prioritization.

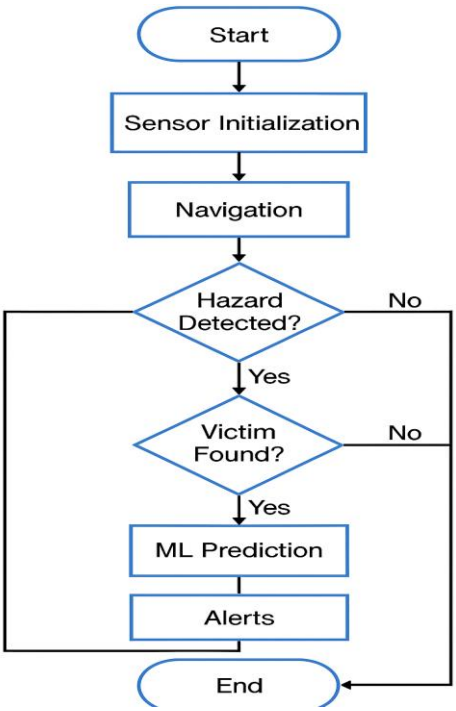

Fig 5.1 Flow Chart of the System

Start

The robot's operation begins when power is supplied to the system. All internal modules prepare for initialization, and the microcontroller becomes active.

Sensor Initialization

In this step, the Arduino initializes all connected components including ultrasonic, fire, gas, metal, temperature, and heartbeat sensors.

- Communication modules (Bluetooth and Zigbee) are also activated.
- The LCD displays the system status to confirm that all modules are responding. This ensures the robot is fully ready for operation before navigation begins.

Autonomous Navigation

The robot now begins moving forward using programmed navigation logic.

- Ultrasonic sensor continuously measures distance from obstacles.
- The microcontroller adjusts DC motor movement using the H-Bridge. This allows the robot to move safely through the environment and automatically avoid obstacles.

Hazard Detection

While moving, the robot actively monitors the environment:

- Fire Sensor → Detects flames or high temperature
- Gas Sensor → Detects harmful gases
- Metal Sensor → Detects metallic debris

  If a hazard is detected, an internal flag is raised, and the robot transitions to the next step.

Victim Check

When hazards or conditions suggest a possible victim, the robot performs a health assessment.

- Heartbeat sensor measures pulse rate
- Temperature sensor records body temperature

These readings help determine whether a trapped person is in stable or critical condition.

ML Prediction

The collected vital signs are transmitted to the remote laptop using Zigbee.

A pre-trained machine learning model analyzes the data and classifies the victim **into:**

- Normal
- Mild Risk
- Critical

  This helps emergency teams identify which victims need immediate attention.

Assist with Arm

If required, the robot uses its pick-and-place arm to support the victim or remove light debris.

- Servo motors or DC motors control precise movements
- Arm can operate autonomously or through Bluetooth

  This adds mechanical support to the rescue operation.

Send Alert

Once all readings and predictions are completed, the robot sends alerts to the control station.

- Alerts include hazard type, victim condition, and sensor readings
- Ensures real-time monitoring and swift rescue decisions

End

The robot completes its workflow.

- It may stop, wait for further instructions, or prepare for shutdown
- Data logs are stored if needed for analysis

This marks the completion of one full rescue cycle.

5.2 Block Diagram

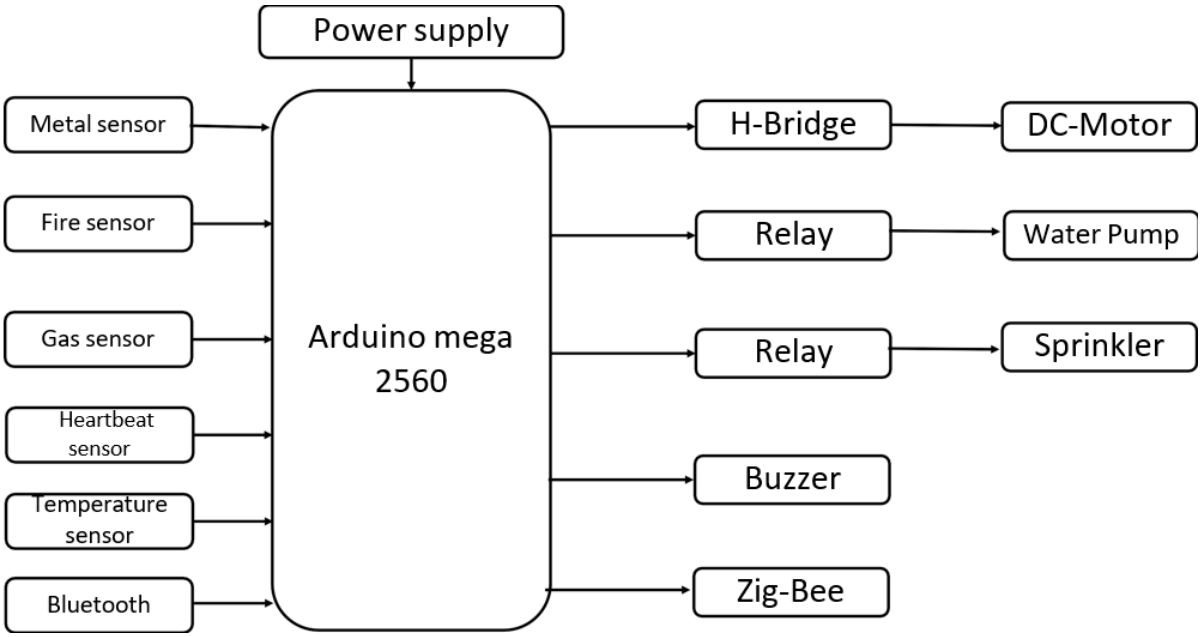

Fig 5.2 Block Diagram

1. Sensor Inputs to Arduino Mega

The robot integrates multiple environmental and biomedical sensors:

- Fire sensor → detects flames or high-temperature radiation.
- Gas sensor → identifies toxic or combustible gases.
- Metal sensor → senses metallic debris or hidden obstacles.
- Ultrasonic sensor → measures distance for autonomous navigation and obstacle avoidance.
- Heartbeat sensor and Temperature sensor → collect victim vitals.

These sensors send analog or digital signals to the Arduino Mega, which continuously reads and processes this data to monitor hazards and victim conditions.

2. Motor Driver (H-Bridge) and DC Motors

The L293D H-Bridge motor driver is used to control the motion of the robot.

- The Arduino sends directional and PWM control signals to the motor driver.
- The motor driver then powers the DC motors, enabling forward, backward, left, and right movement.

3. Pick-and-Place Robotic Arm

The diagram also includes a robotic arm, driven by DC/servo motors through the motor driver.

This arm helps the robot perform light rescue tasks such as:

- Lifting small debris
- Moving obstructions
- Assisting trapped victims

The Arduino controls the arm's movement based on input commands or hazard detection

4. Wireless Communication Units

The block diagram shows two communication channels:

a) Bluetooth Module (HC-05)

Used for short-range control and manual override.

This helps operators control movement or arm actions during testing or emergency intervention.

b) Zigbee Module

Used for long-range data transmission.

The Arduino sends real-time environmental data (fire, gas, metal) and biomedical data (pulse, temperature) to a remote laptop through Zigbee, ensuring secure and reliable communication

5. LCD Display Module

The 16×2 LCD connected to the Arduino displays real-time readings of:

- Sensors
- Victim vitals
- System status
- Alerts

It provides immediate feedback to rescue personnel near the robot.

6. Machine Learning Module (Remote Laptop)

The remote system receives data through Zigbee and processes it using Python.

A machine learning model classifies the victim's condition into:

- Normal
- Mild Risk
- Critical

Chapter Summary

The implementation phase focused on assembling the hardware components and integrating them with the software systems to build a fully functional multitasking rescue robot. All sensors, actuators, and communication modules were connected to the microcontroller, and the required programs were developed to control navigation, sensing, data processing, and real-time communication.

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CHAPTER 6
RESULTS & DISCUSSIONS

CHAPTER 6

RESULTS & DISCUSSIONS

The Results and Discussions section presents the outcomes obtained from the proposed system and explains how these results validate the performance of the model. It highlights the behavior of the system under different conditions, compares the expected and observed outputs, and interprets the significance of the findings. This section also discusses the strengths, limitations, and practical relevance of the results to show how effectively the system meets its objectives.

6.1 Overview of Experimental Setup

The performance of the Multitasking Rescue Robot was tested under controlled and semi-controlled environments for victim detection, environmental monitoring, mobility, and communication efficiency. It comprised an Arduino-based control system, a thermal sensor for temperature detection, a heartbeat sensor for vitals monitoring, a gas sensor for hazardous gas identification, and a wireless camera module that provided real-time visual feedback.

The testing was divided into three stages:

1. Laboratory-level testing of sensors and basic movement.

2. Indoor obstacle testing simulating collapsed structures.

3. Field-level testing on uneven terrain and in low-visibility conditions

Data was collected across multiple trials for its reliability and consistency; data analysis was performed through manual observation and sensor-based recordings.

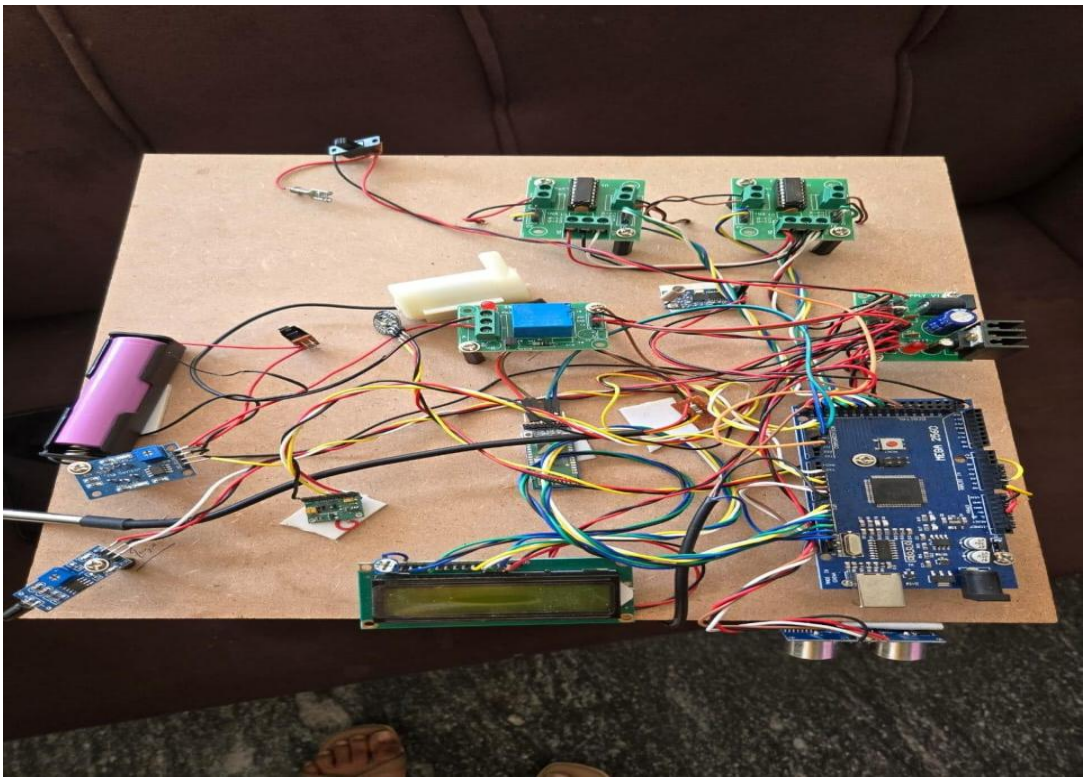

Fig 6.1 Sensor Integration

6.1.1 Sensor Performance Results

The Sensor Performance Results section presents how each sensor in the system responded under different testing conditions. It highlights the accuracy, sensitivity, response time, and reliability of the sensors when detecting the required parameters

Heartbeat Sensor Accuracy

The heartbeat sensor successfully detected pulse variations with an average of 92–95% accuracy in environments with stable lighting and low levels of vibration. When the robot moved on uneven surfaces, vibration interference resulted in slight deviations.

Key Observations:

- Stable readings during stationary states of the robot.
- accurate readings during continuous movement.

Optimal performance was realized when the sensor made firm contact with the test subject.

Temperature Sensor Readings

The body temperature sensor steadily recorded values within the deviation of ±0.3°C, which is acceptable for emergency medical assessment.

Key Observations:

- Accurate detection of fever-range temperatures.
- Rapid sensor response time: ~1.2 seconds.

These environmental sources of heat-lamps, friction-altered the readings somewhat during mobility.

Gas Detection Module

The gas sensor effectively detected the presence of smoke, LPG,and methane based emissions in less than 4–6 seconds.

Key Observations:

- Reliable detection in controlled gas exposure tests.
- Faster response to smoke than methane.

Very sensitive in closed spaces; performance decreases outdoors or with ventilation

6.1.2 Locomotion and Navigation Performance

The Locomotion and Navigation Performance section evaluates how efficiently the robot moves and navigates through its environment**.**

Terrain Adaptability

The robot was tested on different terrains: smooth floors, tiled surfaces, gravel, and low-density debris.

Performance Summary:

- Excellent on flat indoor surfaces.
- Moderate on gravel/dirt paths; some wheel slippage common.
- Against loose debris and uneven rubble, manual control adjustments must be made.

The stability of the chassis and wheel system proved good enough in urban search environments but showed clear shortcomings in rural or heavily collapsed structures.

Obstacle Avoidance and Maneuverability

The robot utilized basic IR-based obstacle detection.

Findings:

- Successful detection for objects >5 cm height.
- Difficulty in identifying very thin or transparent obstacles.
- Turning radius suitable for narrow pathways--approximately 20–25 cm of clearance.
- Obstacle avoidance worked in most environments but lacked the advanced mapping seen in high-end rescue robots.

6.1.3 Testing of Communication and Video Transmission

The Testing of Communication and Video Transmission section examines how reliably the system sends data, control signals, and live video between the robot and the monitoring unit. It focuses on key parameters such as signal strength, transmission delay, video clarity, frame rate, and overall stability of the communication link.

Wireless Camera Module

The real-time camera successfully transmitted live footage within a 10–12 meter indoor range and up to 20 meters outdoors.

Strengths:

- Clear imaging in moderate lighting.
- Minimum delay (<0.5 seconds)

Limitations:

- Noticeable noise in low light.
- Range decreased dramatically with obstacles like walls or metal structures

Bluetooth/Zigbee Communication

Bluetooth communication was effective within 8–10 meters. Zigbee, used where necessary, extended the range and allowed for better penetration through obstacles

Summary Observation:

- Stable control signals, with low latency.
- Temporary loss of signal in metal-enclosed areas.
- Robot remained responsive in most test conditions.

6.1.4 System Integration and Multitasking Efficiency

The System Integration and Multitasking Efficiency section evaluates how well all hardware components, sensors, actuators, and software modules work together as a unified system

Parallel Task Handling

The rescue robot was designed with multitasking in mind to monitor heart rate, temperature, and gas levels while simultaneously providing live video feed.

Experimental findings:

- All sensors worked simultaneously without system crashes
- Minor increase in latency-0.2 to 0.5 seconds during multitasking because of microcontroller load.
- The power consumption increased a lot when all sensors and motors ran together.

Power Management

The robot used a rechargeable battery with an average runtime of 45–60 minutes under full load

Key Points:

- High power was consumed by motor movement.
- Sensor-only mode extended battery life up to 2 hours.
- During rough terrain navigation, the battery drainage rapidly increased.

6.1.5 Performance Analysis and Interpretation

Strengths Identified

- Reliable victim detection by means of temperature and heartbeat sensors.
- Real-time gas detection that increases the safety of the rescue crew.
- Live video feed offering situational awareness.
- Simple and low-cost design makes it apt for early-stage rescue operations.
- Good short-distance maneuverability, indoors especially.

Weaknesses Identified

- Low-light camera performance is somewhat limited.

- Terrain adaptability insufficient for disaster sites that have collapsed heavily.
- Bluetooth range restricts the operational distance.
- Insufficient power supply for extended missions.
- Basic obstacle detection, without advanced mapping or autonomy.

6.2 Practical Implications

These findings suggest that the Multitasking Rescue Robot can be an effective first-response tool in small building fires, indoor accidents, light structural collapses, and many others

However, this model needs to be upgraded for greater mobility and communication in the case of large-scale disasters such as earthquakes or mining collapses.

Fig 6.2 Robotic Arm

Machine learning video : ..\Desktop\ml video.mp4

Chapter Summary

The project successfully developed a multitasking rescue robot capable of sensing hazards, navigating autonomously, monitoring victims, and sending real-time data to rescuers. The integration of multiple sensors, wireless communication, and machine learning improved the robot's efficiency and reliability during rescue operations.

CHAPTER 7
CONCLUSION & FUTURE SCOPE

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The conclusion summarizes the overall performance, outcomes, and effectiveness of the multitasking rescue robot developed in this project. It highlights how the system successfully integrates multiple sensors, communication modules, and autonomous features to assist in rescue operations. This section reflects on the key achievements, the reliability of the prototype, and how well the robot meets the objectives defined at the beginning of the project.

7.1 Conclusion

The Multitasking Rescue Robot developed using the Arduino Mega platform successfully integrates multiple sensors, wireless communication modules, and mechanical systems to provide a versatile and efficient solution for search and rescue operations. By combining environmental hazard detection with victim health monitoring and mechanical assistance, the robot addresses critical challenges faced in emergency scenarios.

Its autonomous navigation capability, enabled by ultrasonic sensors and motor control, allows it to traverse dangerous and complex terrains, minimizing risk to human rescuers. The real-time wireless transmission of sensor data, coupled with a machine learning-based health prediction module, empowers rescue teams to prioritize victims based on their condition, optimizing the allocation of medical resources

.

The modular and cost-effective design ensures scalability and adaptability, making the system suitable for a wide range of applications from disaster management to industrial safety. Overall, this project demonstrates the potential of robotics and intelligent systems to enhance human safety, reduce rescue time, and improve outcomes in life-threatening situations.

7.2 Future Scope

The Multitasking Rescue Robot lays a strong foundation for advanced robotic rescue systems, and there are several avenues for future enhancements to improve its capabilities and applications:

1. **Integration of Advanced Vision Systems:**
   Adding cameras with image processing or thermal imaging sensors can help detect victims visually, identify fire hotspots more accurately, and provide live video feedback to rescue teams, greatly improving situational awareness.

2. **GPS and Mapping Capabilities**:
   Incorporating GPS modules and SLAM (Simultaneous Localization and Mapping)The Multitasking Rescue Robot lays a strong foundation for advanced robotic rescue systems, and there are several avenues for future enhancements to improve its capabilities and applications: algorithms would enable the robot to navigate large-scale disaster areas more efficiently, providing precise location tracking and route planning.
3. **Enhanced AI and Machine Learning Models:**
   More sophisticated AI models can be developed to analyze multi-modal data (such as sound, video, and environmental readings) for better victim detection, hazard prediction, and autonomous decision-making.

4. **Improved Mechanical Design:**
   Developing stronger, more dexterous robotic arms with force feedback and precision control would allow the robot to handle heavier debris and perform delicate rescue operations safely.

5. **Swarm Robotics:**
   Coordinating multiple rescue robots working collaboratively could cover wider areas faster, share data, and assist each other in complex tasks, making rescue efforts more effective.

6. **Battery Life and Power Management:**
   Optimizing power consumption or integrating renewable energy sources such as solar panels can increase operational time during prolonged rescue missions.

7. **Enhanced Communication Systems:**
   Implementing robust communication protocols for reliable data transmission in environments with interference or signal loss would ensure continuous connectivity.

8. **Integration with Emergency Response Systems:**
   Linking the robot's data streams directly to emergency services and healthcare providers can streamline rescue operations and improve response coordination.

Chapter Summary

In the future, the robot can be enhanced with advanced AI algorithms, stronger communication technologies, high-precision sensors, and improved mechanical design to increase accuracy and durability. The system can also be expanded for large-scale disaster scenarios, integration with drones, or deployment in industrial safety environments. These improvements will make the robot more intelligent, adaptable, and suitable for real-world rescue missions.

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